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| **Fiscal Year:** | FY 2007 | **Task Last Updated:** | FY 02/12/2007 |
| **PI Name:** | Dionysiou, Dionysios Ph.D. | | |
| **Project Title:** | High Performance TiO2 Photocatalytic Coatings and Membranes for the Purification, Disinfection and Recycle of Water and Air in Space Applications | | |
| **Division Name:** | Human Research | | |
| **Program/Discipline:** | ADVANCED HUMAN SUPPORT TECHNOLOGIES | | |
| **Program/Discipline--Element/Subdiscipline:** | ADVANCED HUMAN SUPPORT TECHNOLOGIES--Advanced life support | | |
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| **Human Research Program Elements:** | None | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
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| **City:** | Cincinnati | **State:** | OH |
| **Zip Code:** | 45221-0071 | **Congressional District:** | 1 |
| **Comments:** | | | |
| **Project Type:** | Ground | **Solicitation / Funding Source:** | NOT AVAILABLE |
| **Start Date:** | 01/14/2003 | **End Date:** | 01/14/2007 |
| **No. of Post Docs:** | 0 | **No. of PhD Degrees:** | |
| **No. of PhD Candidates:** | 3 | **No. of Master' Degrees:** | 1 |
| **No. of Master's Candidates:** | 1 | **No. of Bachelor's Degrees:** | |
| **No. of Bachelor's Candidates:** | 0 | **Monitoring Center:** | NASA JSC |
| **Contact Monitor:** | | **Contact Phone:** | |
| **Contact Email:** | | | |
| **Flight Program:** | | | |
| **Flight Assignment:** | NOTE: received NCE through 1/14/2007, per PI 5/2006 | | |
| **Key Personnel Changes/Previous PI:** | | | |
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| **Task Description:** | This study deals with the application of novel methods for the fabrication of robust nanostructured TiO2 photocatalytic films for the development of efficient water and air purification systems for advanced life support in space missions and terrestrial applications. The catalytic films immobilized on a variety of ceramic, metallic, and composite substrates will be used to fabricate reactors and membrane systems that will pose unique pore orchestration for the effective capture of incident radiation, enhanced photocatalytic activity, excellent adhesive properties, good mechanical stability, controlled hydrophilicity, and anti-biological fouling capability. These immobilized films will be the crucial component for the fabrication of integrated photocatalytic-disinfection-antifoulant reactors and membranes for the room-temperature production of clean potable and hygienic water in space missions as well as for the purification of water in terrestrial applications. Alternatively, these methods will be used for the fabrication of immobilized TiO2 films aiming at the preparation of surfaces possessing self-cleaning, disinfecting, and anti-biofouling properties for air purification. These purification systems can be used as stand-alone technologies or as supplementary and complementary to existing treatment technologies. In addition, they can serve as back-up processes or as a means of emergency-respond technologies to chemical and biological terrorism.<br><br>We apply a novel fabrication procedure for the preparation of high performance TiO2 films immobilized on non-porous stainless steel substrates. The procedure concerns a novel combination of the alkoxide sol-gel and the colloidal suspension (Degussa P-25) preparation routes for the synthesis of the catalyst. We optimize the P-25 concentration and calcination temperature for enhanced photocatalytic activity and good mechanical stability. The purpose of this research is to better understand the relationship of surface microstructure and film composition and their effect on the photocatalytic activity of the films. After the optimization of the modified sol-gel procedure, we evaluate the photocatalytic activity of these films for the degradation of target contaminants. Our goal is to optimize the conditions for the destruction and mineralization of these compounds with prepared TiO2 photocatalyst and to identify any inhibition factors. Another goal is the design of a new reactor with higher surface area per unit liquid volume treated, increased photocatalytic activity, and better optimization for efficient utilization of UV radiation. In addition, we apply novel nanofabrication and self-assembling procedures to prepare highly porous nanostructured photocatalytic TiO2 particles, films immobilized on nonporous glass substrates, and membranes incorporated onto porous alumina membrane supports. The objective for this task is to use self-assembling modified sol-gel methods for the fabrication of TiO2 thin films of controlled intraparticle (framework) and interparticle (textural) porosity to be used on top of non-porous or porous supports. In the latter case (i.e., porous supports), the method is used for the fabrication of nanocomposite TiO2-ceramic membranes with tailor-designed pore size (mesoporous, 1-50 nm), improved photocatalytic properties, and enhanced mechanical stability.<br><br>This research is considered as an important task within our scope to develop high performance photocatalytic coatings for water purification. The completion of the tasks and objectives is crucial for the fabrication of highly efficient robust integrated advanced oxidation reactors for the purification, disinfection, and recycle of water in space missions. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | The materials prepared in this project and the associated technologies are of interest to the public as means to provide clean water and air and thus to protect human health. In addition, special educational activities focus on public education on issues relevant to nanoscience and nanotechnology. In addition to the development of novel photocatalytic coatings and membranes for water and air treatment, purification and disinfection for Advanced Life Support in space missions, this study can lead to the development of robust photocatalytic water and air purification systems for terrestrial applications. The study can also enhance fundamental knowledge for the fabrication of novel ceramic materials, films, and membranes with anti-biofouling properties. The method for the preparation of films with super-hydrophilic properties can be applied to other support materials with desirable properties for water spreading, light transmittance, and self-cleaning properties. |
| | The proposed work deals with the application of novel aspects on TiO2 photocatalysis for the fabrication of robust nanostructured TiO2 photocatalytic films and membranes. These films and membranes will be an inherent component of efficient water and air purification systems suitable to provide advanced life support in long-duration space missions of NASA as well as in long-term terrestrial applications. These purification systems can be used as stand-alone technologies or as supplementary and complementary to existing treatment technologies. In addition, they can serve as back-up processes or as a means of emergency-respond technologies to chemical and biological terrorism.<br>During the project period, we investigated novel synthesis routes and the associated fundamental mechanisms for the preparation of highly active, tailor-designed, and mechanically stable immobilized TiO2 thin films and mesoporous membranes on top of ceramic and metallic supports. These films and membranes will be used for the fabrication of tubular photocatalytic non-membrane reactors and integrated disinfection-photocatalytic-membrane reactors for the purification and disinfection of water and air in space applications. In summary, the specific scope and objectives of this four-year project were: (1) To fabricate immobilized to TiO2 photocatalytic films on top of non-porous supports using a novel combination of alkoxide sol-gel methods and the colloidal suspension preparation route. These films should have optimum interparticle porosity (textural) for the effective capture of UV radiation, high surface area, increased photocatalytic activity and enhanced mechanical stability for durable performance in long-duration operations, (2) To fabricate immobilized TiO2 nanostructured photocatalytic thin films on top of non-porous supports via self-assembling sol-gel methods. In addition to the films described above, these films also pose intraparticle porosity (framework) and thus higher surface area. They are also of much smaller film thickness. The formation of the intraparticle porosity is achieved using sol-gel methods modified by the addition of novel directing agents (i.e., templates) to form precise geometric micellar assemblies. After dip-coating and heat/treatment steps, the resulting TiO2 films exhibit intraparticle mesoporosity which can mimic the size of the micellar assemblies, (3) To develop novel supported TiO2 nanostructured photocatalytic membranes with controlled mesoporosity (1-50 nm) and hierarchically orchestrated ceramic framework, high photoactivity, thin film thickness (<200 nm), and enhanced mechanical stability. These membranes are prepared using self-assembling sol-gel methods to coat TiO2 porous films on top of porous supports. Such membranes should have good permeation properties, increased selectivities, and photocatalytic action for simultaneous treatment and disinfection of water, and (4) To study the associated physicochemical phenomena and investigate the fundamental role of various experimental process parameters on the properties of the final TiO2 films and membranes during the fabrication procedures. These parameters include the properties and concentration of surfactant (i.e., properties and size of micelles), the dip-coating procedure (i.e., sol withdrawal speed), and the thermal treatment conditions (i.e., calcination temperature). The study also aims at determining the conditions that result in |

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| **Task Progress:** | absence of catalyst fouling and membrane cracking.<br><br>In order to achieve these, we applied a novel fabrication procedure for the preparation of high performance TiO2 films immobilized on non-porous substrates. The procedure concerns a novel combination of the alkoxide sol-gel and the colloidal suspension preparation routes for the synthesis of the catalyst. This concerns the incorporation of highly active TiO2 nanoparticles (Degussa P-25) in the sol prior to dip coating of a support material. The coatings are then subjected to drying and heat treatment to obtain the desirable crystal phase of the catalyst. Following this titania powder-modified sol-gel procedure, TiO2 films have been immobilized on 304 stainless steel substrates. Various instruments and techniques were used to characterize the coatings and TiO2 materials and the photocatalytic activity of the TiO2 films was evaluated using 4-chlorobenzoic acid, phenol, and creatinine. During the photocatalytic degradation of creatinine (a urine metabolite), we investigated its degradation pathway.<br><br>In addition, we investigated the synthesis, properties and photocatalytic activity of porous TiO2 films and membranes using self-assembling methods. In this work, we explored an interesting nanotechnological method to fabricate highly mesoporous materials with extremely high surface area, desired pore structure and size, and catalytic activity. This method was based on self-assembling of amphiphilic surfactants that form micelles and assist the titanium alkoxide precursors to self-organize in the micellar corona. After calcination where the organics sol precursors are pyrolyzed, the pore size of the TiO2 catalyst can mimic the size of the micelle. In this study, mesoporous TiO2 particles, films, and membranes were synthesized using this method. As a result, the use of self-assembling surfactants in titania sol resulted in promising TiO2 materials with better structural properties including homogeneity, controlled pore size and distribution, high water permeability, high organic retention, and enhanced catalytic properties (high surface area, active anatase phase, small crystal size, and high material crystallinity). This sol-gel method modified by varying the type and concentration of the surfactants can be useful to prepare nanostructured anatase TiO2 particles, thin films and membranes with high photocatalytic activity and desired pore structure for environmental applications and to synthesize similar crystal nanostructures of other oxide materials for applications in catalysis and separations.<br><br>In general, the results obtained in this work are promising and are expected to be of high interest to NASA, the scientific community, and many relevant industries for the development of sustainable photocatalytic reactors as well as for the development of coatings with task-specific applications. The good structural properties and enhanced catalytic activities of the prepared TiO2 materials are important requirements for the fabrication of sustainable photocatalytic reactors for water and air purification. The tunable hydrophilic properties of these coatings are also of interest to other industries dealing with special glass fabrication, automotive industry (i.e., completely wetable surfaces), indoor air disinfection and purification.<br><br>In addition to physical research activities, the project provided valuable opportunities to graduate, undergraduate, and high school students. The students had an opportunity to learn novel procedures for the fabrication of nanostructured TiO2 films and membranes and exchange valuable experience and knowledge in the area of materials synthesis and their environmental applications. The students involved in this project also had an opportunity to develop their analytical skills for the detection of primary compounds and their secondary metabolites. In addition, they enhanced their understanding for the reactivity of hydroxyl radicals and the degradation routes that these active species follow. Students in the Environmental Science and Engineering program at the University of Cincinnati also had the opportunity to learn current issues in the field of nanotechnology and became familiar with the synthesis, characterization, and environmental applications and implications of nanomaterials. Dr. Dionysiou and his group also promoted awareness on the health threat of environmental pollution in water and wastewater as well as the importance of environmental preservation to people, including public and environmental affairs around the Cincinnati regional area. In order to disseminate the importance of nanotechnology in the area of environmental remediation to colleagues, public, and communities, Dr. Dionysiou and his students presented their research findings on the degradation of hazardous organic contaminants using TiO2 photocatalytic films and membranes at several regional, national, and international conferences and meetings and they were acknowledged with several awards, fellowships, and scholarships. Finally, Dr. Dionysiou's collaboration with other national and international universities, and the US Environmental Protection Agency expanded the opportunities for advanced and international level research. |
| **Bibliography Type:** | Description: (Last Updated: 10/14/2009) |
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| **Awards** | Chen Y. "2006 Summer Graduate Student Research Fellowship (University of Cincinnati Research Council), July 2006." Jul-2006 |
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